

Zhuoning (Johnny) XU

Jersey City, NJ | 551-998-6205 | zx2094@nyu.edu | www.linkedin.com/in/zhuoning-xu-johnny | https://github.com/KeiraXu03

EDUCATION

New York University

New York, NY

MS, Computer Science (Courant Institute of Mathematical Sciences, GPA: 3.78/4.0)

Sept. 2025 - May 2027 (expected)

- **Coursework:** Programming Language (Racket), Fundamental Algorithms, Mathematical Techniques for CS Applications, Machine Learning (Spring 2026), Operating Systems (Spring 2026), Natural Language Processing (Spring 2026)

The Hong Kong Polytechnic University

Hong Kong

BEng, Major in Electronic & Information Engineering; Minor in Computing (GPA: 3.8, Top 5%)

Sept. 2021 - Jun. 2025

- **Honors:** Dean's Honors List (2021-2024), First-class honours degree
- **Coursework:** Computer System Fundamentals (**x86 Assembly**), Programming Fundamentals (**C/C++**), Object-Oriented Programming (**Java**), Web Application Design and Development (**Flask**), Operating Systems, Database System (**SQL&NoSQL**), Data and Computer Communication, Data Structures, Computer Vision and Pattern Recognition

SKILLS

- **Programming Languages:** Assembly Language, C/C++, **Java**, Web programming (PHP, HTML, CSS, JavaScript, AJAX), **Python** (Pandas, NumPy, Scikit-Learn, Matplotlib, **PyTorch**), SQL
- **Development Tools:** MySQL Workbench, Matlab, Linux Shell, PyCharm, Docker, Android Studio, MS Office Suite, Git

INTERNSHIP EXPERIENCES

Assistant Software Engineer Intern - Front-end & Data Monitoring, O&M Software Department

Shenzhen, China

YOTAI Digital Energy Technology Co., Ltd

Jun. 2024 - Aug. 2024

- Resolved data loss, errors, and UI lag by refining Umi/**Node.js** and shipping reusable hooks (**WebSocket**, **device model**, **device data**); improved page responsiveness by 15%
- Built the monitoring web app and a unified equipment model, standardizing backend APIs and stabilizing live views with an efficient subscription and update pipeline
- Developed a **Python analysis module** to automate reporting, boosting metric accuracy, and cutting preparation time

Data Engineer Intern - Business Intelligence & Visualization, Technical Department

Hangzhou, China

Hangzhou Green Cloud S & T Co., Ltd.

Jun. 2023 - Jul. 2023

- Used **SQL** to collect, clean, and combine operational data so dashboards had trustworthy inputs and analysts spent less time on one-off data requests.
- Led desktop and **mobile report development** for 10+ clients; built **data visualization** platforms with FineBI and launched command-center dashboards and large-screen displays with clear, interactive designs.
- Captured requirements and streamlined processes to design custom reports; applied **statistical analysis** to deliver forecasts and recommendations that improved operational efficiency

Quantitative Trading Internship

Hong Kong

MOL Quant Limited

Jan. 2025 - Jun. 2025

- Built a **Python-based** quantitative trading framework from scratch, covering data ingestion/cleaning, vectorized data processing, signal-research modules, execution interfaces, and a fully vectorized backtesting engine.
- Established an AI Agent platform to conduct real-time sentiment and event analysis on financial news, generating investment recommendations and risk alerts to provide data support for intraday strategies and stock selection decisions.

RESEARCH EXPERIENCES

A Multi-Perceptual Learning Network for Retina OCT Image Denoising and Classification

Jan. 2024 - Sept. 2024

- Replaced VGG with **ResNet** and integrated **Multi-Perception Learning** with a Triple-Cross-Fusion GAN (**PyTorch/OpenCV**) for joint denoising and classification
- **Improves image quality and increases classification accuracy** by over 1% compared with the baseline on the same dataset, while maintaining clinician-friendly real-time inference on GPU

Transformer-based Vision-Language Jigsaw Puzzle Assembly (Accepted by AAAI-26 Workshop)

May. 2025 - Sept. 2025

- Introduced textual descriptions as global semantic anchors to resolve patch-level ambiguity; extracted visual and text features via ViT and BLIP, aligning cross-modal representations at both patch and image granularities using InfoNCE contrastive loss, with Hungarian matching for optimal permutation assignment.
- Led the full research pipeline including ablation studies over local/global alignment branches; the proposed method achieves significant permutation accuracy gains over pure-vision baselines, validating the role of language semantics in disambiguating visually similar fragments.

Personalized Online RL System for User Style Adaptation via GRPO

March 2026 - Present

- To address the inability of existing agent systems to retain users' problem-solving habits, we apply online continual fine-tuning to Qwen3.5-4B under the OpenClaw-RL framework, using the user's subsequent message as a natural reward signal and a PRM to score each response turn. The auxiliary model progressively learns the user's stylistic and task-handling preferences, then generates personalized behavioral specifications at inference time, injected as prompts into the primary model — replacing static skill libraries and memory-retrieval pipelines. Ongoing.

PUBLICATIONS

- "A Multi-Perceptual Learning Network for Retina OCT Image Denoising and Classification" 2024 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (DOI: 10.1109/apsipaasc63619.2025.10848686)
- "MFGAN: OCT image super-resolution and enhancement with blind degradation and multi-frame fusion" International Workshop on Advanced Imaging Technology (IWAIT) 2025 (DOI: 10.1117/12.3057230)